A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 24

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Occupancy sensors detect infrared / movement and switch lights or appliances off when no one is in the room **1** Programmable thermostats set different temperatures for different rooms and times so unused rooms / periods are not heated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** LEDs convert a much larger proportion of electricity to light rather than heat **up to 2** Washing machines with cold / low-temperature cycles and fast spins (less drying) – dishwashers using less water so less heating – cookers with double / triple glazed doors – fridges with linear compressors – LED screens using half the power of plasma – germanium transistors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Turning off lights / appliances rather than standby **1** Turning thermostats and radiator valves down / off in unused rooms **1** Only heating the water needed (kettle) **1** Showering instead of bathing – eco-cycles **1** Choosing smaller vehicles / less travel **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Stored hot water loses heat continuously through the tank walls and pipes **1** Heating as needed (or just before) avoids these standing losses – if storage is needed the tank should be well insulated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 232-233; AQA spec 3.3.4.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Net held with its straight edge on the river bed facing the current (open to the flow) **1** The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) **1** Same substrate type, water depth and flow rate at both sites – same net mesh – repeat samples – invertebrates identified with a key and counted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	3
02.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Difficult to standardise the vigour / area kicked **1** Buried organisms or those fixed to stones are not dislodged **1** Some organisms swim and avoid the net **1** Flow rate varies and affects capture **1** Fragile organisms destroyed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed area of bed is enclosed by a frame so the sampled area is standardised – side nets reduce escapes **1** But only works in fast enough flow and is hard to set on rough substrates / deep water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event **1** Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** UV light splits diatomic oxygen (O2) into two monatomic oxygen atoms (O) **1** Monatomic oxygen reacts with diatomic oxygen to form ozone (O3) **1** UV light also splits ozone into O2 and O – the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Useful:** boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants) – dissolve grease but do not damage electrical components (solvents) – non-flammable – non-toxic **up to 2 Persistence:** chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere **up to 2 Dissociation by UV:** in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals **up to 2** Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation – chlorine is regenerated so one atom destroys many ozone molecules – bromine / iodine act similarly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **Persistence** → does not break down / stays in the environment for a long time **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 3 points below. **1** UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) **1** Damage to leaf tissue and reduced photosynthesis / crop yields (plants) **1** Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.4	1 mark for the point below. **1** An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
03.5	1 mark for the point below. **1** Concentration varies with altitude and the stratosphere is not normally sampled – Dobson Units measure the total column thickness instead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
03.6	1 mark for the point below. **1** The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions **1** The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.2	**1 mark each** 1 mark for each of the 2 points below. **1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) **1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	1 mark for the point below. **1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its total annual discharge	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Public supplies cannot match consumer demand / the water is often cleaner than rivers **1** It also reduces urban flooding after heavy rain – surplus can be used for aquifer recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** The reservoir is created in the estuary so no farmland / settlements need to be flooded **1** But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1 Production prevention:** desulfurisation of fuels before combustion – low-temperature (fluidised bed) combustion to prevent NOx – smokeless fuels **1 Prevention of release:** electrostatic precipitators / bag filters / cyclones for smoke – catalytic converters – flue gas desulfurisation – bund walls – landfill liners **1 Post-release remediation:** oil spill clean-up (booms, skimmers, dispersants, bioremediation) – phytoremediation of metal-contaminated land – liming of acidified lakes **1 Alternative processes:** electric vehicles instead of petrol / diesel – renewable energy instead of fossil fuels – pyrethroids instead of organochlorines – fuel cells instead of internal combustion engines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	4
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Remediation is often incomplete / cannot recover all the pollutant (eg oil spills, dispersed pollutants), so some damage still occurs **1** Prevention avoids the damage entirely and is usually cheaper than clean-up / avoids polluter-pays costs – moving from post-production treatment to non-release is a circular economy principle **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 4 points below. **1** Satellites detect and track the extent / movement of oil slicks over large areas, quickly and repeatedly, so clean-up can be directed and polluters identified **1** GPS tracking monitors ship positions so navigation errors / collisions / groundings are reduced and offshore shipping routes are enforced **1** Polymers with a high surface area adsorb metal ions from contaminated water / effluent so they can be removed and recovered **1** Plants that absorb heavy metals (hyperaccumulators) are grown on contaminated land, harvested and disposed of / metals recovered, cleaning the soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (6400 - 220) / 6400 x 100 **1** = 96.6% (accept 97%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Switch from coal to natural gas (which is desulfurised) / closure of coal-fired power stations removed the main SO2 source **1** Flue gas desulfurisation fitted to remaining coal plants – coal / oil desulfurisation **1** NOx comes mainly from road transport and vehicle numbers / distance travelled increased greatly **1** NOx is produced from air (nitrogen + oxygen) in any high-temperature combustion, so cannot be removed from the fuel like sulfur **1** Catalytic converters only work when warm / diesel engines produce more NOx / older vehicles remained in use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Ozone and PANs are secondary pollutants formed by photochemical reactions involving NO2 (broken down by UV to monatomic oxygen which forms ozone) **1** If the primary pollutant is reduced the reactants are not present so the secondary pollutants cannot form **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** Lichens are very sensitive to acidic conditions / SO2 and different species have different tolerances **1** The species present, their abundance, size / health can be used as a biotic index of the pollution level **1** They give a long-term integrated measure of pollution rather than a single instantaneous reading – cheap, no equipment needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-253; AQA spec 3.4.3.2.1-3.4.3.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10^3 / 6^3 = 1000 / 216 **1** = 4.6 times (accept 4.63) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
08.2	**1 mark each** 1 mark for each of the 5 points below. **1** C has the highest velocity (most energy, and energy rises with the cube of velocity), strong reliable winds over water, few land-use conflicts, but long expensive cable and offshore installation / maintenance costs **1** A is windy but on a migration route (bird strikes) and peat bog (foundations / tracks damage hydrology; carbon release) – far from grid – scenic objections **1** B is close to the grid with good wind but near a village (noise, visual objection, shadow flicker) - possible with careful siting **1** D is cheap to connect and land can still be farmed between turbines, but low velocity gives little energy **1** Judgement, eg C or B – turbines spaced 3-5 blade diameters apart to avoid wind shadow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	5

Do not write outside the box

Question	Marking guidance	Marks
08.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Driven by wind from any direction so no motor to turn them – suitable for turbulent urban sites **1** Generator can be at ground level so a lighter tower **1** Quieter **1** Operate at lower wind velocities **1** Blades more visible so fewer bird deaths **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
08.4	1 mark for the point below. **1 No gearbox:** fewer breakdowns, less friction so start generating at lower wind speeds, quieter, lighter nacelle (DC converted to AC by an inverter) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 206-209; AQA spec 3.3.4.1.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1 LEO:** high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover **1** But each area is only revisited every 15 days so they cannot follow rapidly changing weather **1 Geostationary:** constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1** Wind velocity **1** Ocean currents / sea surface temperature **1** Wave height **1** Ice thickness **1** Vegetation cover **1** Sea level **1** Atmospheric temperature / composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome **1** If the prediction matches, the model can be trusted (with caution) – it is continually modified as more data are collected and differences analysed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Models need data from all regions / oceans / altitudes – gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions **1** Unrepresentative data (eg urban heat islands, short time series) would bias the model – remote sensing helps fill gaps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable – demand fluctuates daily, weekly, seasonally and short-term – supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants – large-scale electricity storage not yet possible directly so energy must be converted • **Pumped-storage HEP:** proven, rapid response, peak shaving, but limited sites • **Batteries:** lithium-ion for grid and vehicles – factors affecting viability – V2G • **Hydrogen:** electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid – high energy density enables vehicle use • Compressed air with heat recovery – thermal storage (molten salt for CSP, high-volume stores, thermal mass) – flywheels – supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage – energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **HEP** → electricity generated from falling water • **energy conservation** → using less energy, eg by insulation or behaviour change **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (9 marks)	
	Renewable electricity is intermittent - solar stops at night, wind on calm days, tidal power at slack water - and some of it is unpredictable, while demand fluctuates by the minute, day, week and season, so supply and demand rarely coincide. Without storage, surplus renewable output is wasted and shortfalls must be met by fossil-fuel peaking plants, which defeats the purpose. Electricity itself cannot be stored at scale, so the energy must be converted into another form and back.	
	Pumped-storage HEP is proven and responds within seconds for peak shaving, but suitable sites are few. Lithium-ion batteries now work at grid scale and in vehicles, with vehicle-to-grid schemes turning car batteries into distributed storage, though lithium supplies and lifetime limit them. Hydrogen from electrolysis can be stored, used in fuel cells or injected into the gas grid (power-to-gas), and its high energy density suits vehicles, although each conversion loses energy. Compressed air with heat recovery, molten-salt thermal stores for concentrating solar plants, thermal mass in buildings, flywheels and supercapacitors each fit particular timescales.	
	Storage is not the only answer: a diverse mix of wind, solar, tidal and biofuel smooths supply, smart-grid management shifts demand, and conservation reduces peaks. But these reduce rather than remove the mismatch. Short-term storage can be batteries and pumped hydro; the harder problem is seasonal storage, where only hydrogen or fuels made from surplus electricity look plausible. Without both, renewables can supplement but not dominate a country's supply.	
	Weaker answer - Level 2 (5 marks)	
	Renewable energy is intermittent. Solar panels do not work at night and wind turbines do not work when it is not windy, but people still need electricity then. Storage means the energy can be saved for when it is needed. Pumped storage pumps water uphill when there is spare electricity and lets it flow back through turbines at peak times. Batteries can also store electricity. Hydrogen can be made from electricity and used in fuel cells. So storage is needed for renewables to provide most electricity.	
	What makes the difference	
	• Level 2 explains the mismatch and describes three storage methods correctly, but does not evaluate them (sites, lithium, conversion losses, timescales).	
	• Level 3 distinguishes short-term from seasonal storage - the key insight for 'most of a country's electricity'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 also considers the alternatives to storage (diversity, grid management, conservation) and explains why they are insufficient alone.	
10.2	1 mark for the point below. **1** Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Problematic properties of solar:** intermittency, unreliability from cloud, low energy density, changing angle – locational constraints (deserts, latitude) • **Harnessing solar:** photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms – environmental impacts of manufacture and land use • **New solar technologies:** multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • **HEP:** gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links – combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Historical data limitations – proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • **Satellite sensors:** visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring – passing through cloud • **Sensor carriers:** ROVs / drones, AUVs, balloons, aircraft, animals – data recorders for abiotic factors – GPS tracking of floats and sampling sites – Argo floats and ocean monitoring • Climate models and their assumptions – monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • **Evaluation:** improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	

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